Optimal play in the tile stacking game High Tide

Gweneth Andersen

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G. Andersen is with Department of Physics and Applied Physics, University of Massachusetts Lowell, Lowell, MA, USA and Department of Physics, Northeastern University, Boston, MA, USA. Corresponding author: Gweneth Andersen (e-mail: andersen.g@northeastern.edu).

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***Abstract*—We show that the game graph of the tile stacking game High Tide is finite, directed, and acyclic, and exploit these properties and the board's geometric symmetries to prove that optimal play is, in principle, computable by dynamic programming. Generalizing the game rules in an evolutionary graph theory framework to define the "N-Tide" family, we compute a minimax tablebase of win probability-maximizing paths on the game graph of a reduced version called 2-Tide. For 2-Tide, ε-greedy play simulations reveal a convergence to 100% first-player win rate for 7 out of 11 canonical initial board positions and 100% second-player win rate for the remaining 4 when both players select moves that maximize their win probability. Using this insight, features of optimal paths in 2-Tide, and a Monte Carlo Tree Search (MCTS)-based proxy for optimal play in 3-Tide, we derive strategic heuristics and identify avoidable conditions that can lead to vulnerable positions.**

***Index Terms*—Tile stacking games, Dynamic programming (DP), Monte Carlo Tree Search (MCTS).**

I. Introduction

HIGH TIDE is a two-player abstract strategy board game created by tabletop game designer and developer Marceline Leiman and published in 2025 by Underdog Games. Its simple, yet unique rule set governing movement introduces new problems in area control strategy, and bears a resemblance to the competition arising between two species in a mutualistic relationship challenged by non-destructive environmental shift. In this ecological analogy, the objective represented by High Tide's rule set is for a species to preserve its range in a shared habitat with dwindling habitable space. Formalizing High Tide as a novel dynamical system may lend insight to the study of similar phenomena among real-world species affected by climate change. This will be the object of future work.

Here, we formalize the family of dynamical systems prescribed by High Tide's movement rules on an evolutionary graph with arbitrary topology, and approach the game as a special case with initial conditions whose convenient symmetries make possible a comprehensive analysis of optimal play by computer simulations. We prove the acyclicity of the game graph, define a generalized form of the original game called N-Tide, and compute exact minimax values for initial configurations of a simpler variant, 2-Tide. In doing so, we reiterate the practice of "micro-reduction" on hexagonal lattices used by Craige Schensted and Charles Titus in [1] as well as Steven Meyers and Jack van Rijswijck in [2] for the connection games Hex (invented by Piet Hein and popularized by John Nash) and Y (shown by Schensted and Titus to generalize Hex) respectively. While High Tide is not a connection game in the sense of path formation being an intrinsic objective, as in Hex and Y, we find in it new applications of mathematical results—such as Nash's strategy-stealing argument, see [3]—derived from games in this genre.

II. Game Definition

A. Basic Rules

The game board consists of 19 hexagonal tiles—7 pink (**P**), 7 blue (**B**), and 5 yellow (**Y**)—initially arranged in a hex "flower" configuration with one yellow tile at the center, a first ring of petal tiles in a random permutation of **PPBBYY** (or **2P2B2Y**), and a second ring in a random permutation of **5P5B2Y**. An example configuration is shown in Figure 1A. Hex cells are labeled with compact axial coordinates (q, r) in the efficient Array Set Addressing scheme, described extensively in [4] and elaborated in the supplementary material.

Players take turns moving tiles of their own color (pink or blue) or neutral tiles (yellow) on top of other adjacent tiles, forming stacks. Here, the board will be encoded by cell indices v and the height (number of tiles) h_v of the stack at cell v, where $h_v = 1 \; \forall \; v$ in any initial configuration. Movement must satisfy four constraints.

First, only the apical (topmost) tile of a stack may be moved. Second, tiles must only move upward: a tile in a source cell v_i may be moved to a target cell v_j if and only if its new height would exceed its current height: $h_j \geq h_i$. Third and fourth are the topological constraints, which are best expressed by treating the board as a graph $G(V, E)$ whose vertices $v \in V$ are cells and whose edges E denote adjacent pairs of cells. There are no self-edges. Because each vertex v_i is associated with a pair of coordinates (q_i, r_i), an edge e_{ij} exists between v_i and any vertex v_j with coordinates equal to the vector sum of (q_i, r_i) and one of the ASA basis vectors $\hat{e} \in \{(1,0),(0,1),(-1,0),(0,-1),(1,-1),(-1,1)\}$.

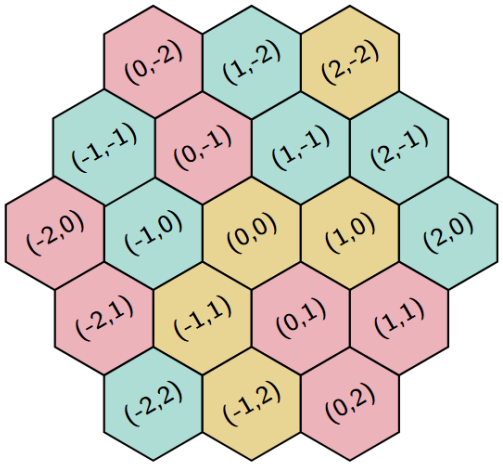
$$E = \{\{v_i, v_j\} \in V \mid i \neq j, \; \exists (a,b) \in \hat{e} \\ \text{such that } q_j = q_i + a \text{ and } r_j = r_i + b\} \tag{1}$$

Legal moves must leave the whole set of occupied cells a connected component, and only tiles in cells with at least two unoccupied adjacent cells (i.e., source cells with adjacency degree $k_v \leq 4$) may be moved. A player must make a legal move each turn if able. Figure 1B shows an example of a legal move.

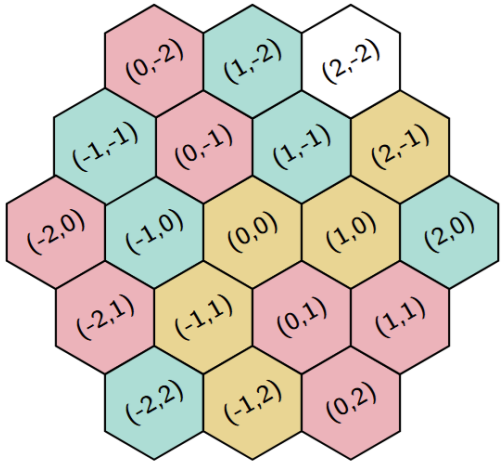
Termination occurs when the player can make no legal moves (including moving neutral tiles) on their turn. The winner is the player with the greatest number of apical tiles of their color; in the event of a tie, the winner is the player whose highest apical tile is at the greater height. If there is full parity of apical tile heights, then the final tiebreak condition awards victory to the player who moved last.

B. Formal Properties

High Tide is a partisan, finite, zero-sum, perfect information abstract strategy game. We will show that its properties of directedness, acyclicity (DAG), and boundedness permit, in principle, the determination of an optimal play policy by dynamic programming (DP).



(a) An initial board position.



(b) Position after a first move.

Fig. 1. Board positions in High Tide with ASA coordinates. Figure (b) shows the movement of the neutral tile from (2,-2) onto the blue tile at (2,-1).

Lemma 1. *The game graph Γ of High Tide is a directed graph.*

Proof. Given a game position s encoding the full configuration of tiles (i.e., each tile's cell index v and stack position h, including hidden tiles) and the player p whose turn it is, the set of legal moves is $\mathcal{M}(s,p)$. A legal move $m \in \mathcal{M}(s,p)$ is a mapping of the position s to a successor position s', $m : s \to s'$. The vertices of Γ occupy the set of all reachable positions $s \in S$, and the set of edges M is the union of all move sets $\mathcal{M}(s,p) \cup \mathcal{M}(s,q)$ for players p and q.

Because of the upward mobility constraint, legal moves do not have legal inverses $m^{-1} : s' \to s$. This makes Γ directed. Formally, this may be stated:

$$\forall m \in \mathcal{M}(s,p) \mid m : s \to s',\ m^{-1} \notin \mathcal{M}(s',q) \tag{2}$$

□

To express the other consequences of the movement constraints in High Tide algebraically, we define a convenient strict potential function $\phi(v)$ equaling the h_v-th triangular number:

$$\phi(v) = \frac{h_v(h_v+1)}{2} = T(h_v) = 1 + 2 + \ldots + h_v \tag{3}$$

Lemma 2. *The global potential $\Phi = \sum_v \phi(v)$ strictly increases with each move and has a finite upper bound.*

Proof. Moving a tile from v_i to v_j results in new stack heights $h_i - 1 \geq 0$ and $h_j + 1$, and thus new cell potentials $\phi'(v_i) = T(h_i - 1)$ and $\phi'(v_j) = T(h_j + 1)$. Consequently, each legal move changes Φ by an amount $\Delta\Phi$:

$$\Delta\Phi = \Big(T(h_i - 1) - T(h_i)\Big) + \Big(T(h_j + 1) - T(h_j)\Big) \tag{4}$$

$$T(h_i - 1) - T(h_i) = -h_i \tag{5}$$

$$T(h_j + 1) - T(h_j) = h_j + 1 \tag{6}$$

$$\Delta\Phi = h_j + 1 - h_i \tag{7}$$

The upward mobility constraint demands that $h_j \geq h_i$, so it follows that $\Delta\Phi \geq 1$ for all legal moves. Therefore, the global potential increases monotonically over the course of the game, and is bounded at a maximum value of $\Phi_{max} = T(19) = \frac{19 \cdot 20}{2} = 190$ which may be reached only in the subset of terminal positions where all 19 tiles occupy a stack in one cell. A *terminal position* is defined as one with an empty move set, $\mathcal{M}(s,p) = \emptyset$. □

Lemma 3. *The game graph Γ of High Tide has no directed cycles.*

Proof. Because a player must choose a legal move each turn if able, there are no self-loops on Γ.

$$\nexists\, m \in \mathcal{M}(s,p) \mid m : s \to s \tag{8}$$

Furthermore, for a directed cycle to exist among $n \geq 2$ positions in Γ, it would be necessary for Φ to increase with every move on the cycle; but Φ is bounded, so this is not possible. □

III. Generalization

A. Micro-Tides

Disregarding tile color and stack order, the number of weak compositions of 19 tiles across 19 cells and 19 stack positions equals $\binom{19+19-1}{19} = \binom{37}{19} = 17,672,631,900 \sim 10^{10}$ positions. Color distributions in the first and second ring confine the subset of legal initial positions to $\left(\frac{12!}{5!5!2!}\right) \cdot \left(\frac{6!}{2!2!2!}\right) = 16,632 \cdot 90 = 1,496,880$. The movement restriction rules significantly reduce reachability, and the board's convenient

D6 dihedral symmetry can be exploited to further curtail enumeration since many positions are equivalent under rotation or reflection. Precise counting of valid positions and paths remains challenging for the published version of the game; however, we may generalize its structure and analyze simpler variants with the same rule set to derive some insight for improving play of the original High Tide.

The simplest classes of boards may be generated in a manner similar to the Y-reduction procedure in [1]. While the board comprising a single tile of any color is non-instructive due to its lack of movement, the two-tile board in any combination of colors (**1P1B**, **1P1Y**, **1B1Y**) provides a trivial case with a single move. The player who makes this first and only move, P_1, is the winner, regardless of board composition. This is because even if the board lacks any tile of their color, it must then still include one neutral tile, which they can move, and by the last-mover tie-breaking condition, that player wins.

Slightly more interesting are the three-tile boards in the class **1P1B1Y**, of which there are three. First is the line case where the center tile is neutral, and thus immovable. This is also trivial, as each player has only one legal move, although the second player, P_2, necessarily wins. If the neutral tile occupies a peripheral position, the player owning the other peripheral tile has two legal moves if they play first; only one of these moves, the one in which they move the neutral tile and deprive their opponent of any moves, leads to victory. The player owning the center tile cannot win. However, the triangular case produces an interesting symmetry: P_1 initially has four available moves, among which only one leads to victory—they must move the neutral tile onto their opponent's tile. The three-tile boards are useful precisely because they illustrate the importance of neutral tiles in basic strategy, and their game graphs are easy to draw.

This principle is further reinforced by the four-tile boards. While hundreds technically exist, with 3^4 color configurations and 7 distinct topologies, only boards containing at least one blue and one pink tile are meaningful, and only four, all 2×2 grids, are compared here. The first two have two pink and two blue tiles—**2P2B**—with either one (A) or both players (B) owning adjacent tiles. The latter two include two neutral tiles—**1P1B2Y**—and either have both player-owned tiles mutually adjacent and neutral tiles mutually adjacent (C) or have player-owned tiles *not* mutually adjacent but neutral tiles still mutually adjacent (D). All four of these boards exhibit some symmetry, but have different outcomes that depend on the number of neutral tiles and the adjacency of opposing tiles, both properties that happen to provide useful information for predicting the outcomes of games on more complex boards in the genre.

In (A), the player whose tiles are adjacent to each other (P_{adj}) has 6 legal moves if they start, whereas their opponent (P_{nadj}) has only 4 legal moves if they start. However, if P_{adj} starts, they can never win. On the other hand, all of P_{nadj}'s starting moves are virtually identical, and whether or not they win depends entirely on P_{adj}'s decision to move their sole uncovered tile onto the tile their opponent did not move. If P_{adj} selects this move instead of moving onto the budding stack, they will force P_{nadj} to cover the tile that just moved, releasing P_{adj}'s first covered tile, which then victoriously climbs the new stack. Otherwise, a similar sequence plays out but ends in victory for P_{nadj}.

In (B), P_1 has 5 moves: two moves involve covering one of their own tiles, with one leading to a loss and the other, by forcing P_2 to cover their own tile, leading to a color-inverted win; and of the remaining three, two similarly force P_2 into a losing move, and the final one immediately ends the game by invoking the connectivity constraint.

The presence of neutral tiles in (C) and (D) reveals the potential utility in using them to cover one's own tiles, but also the dependence of this utility on the particular topology of the *intersecting* subgraphs consisting of the tiles movable by each player on boards of the same shape. In (C), where player-owned tiles are mutually adjacent but only one player's tiles border both neutral tiles, that player's only hope for a win after making the first move is through using the neutral tile non-adjacent to P_2's tile to cover the other neutral tile and having P_2 commit a blunder by proceeding to cover the neutral stack with their own. In all other outcomes, P_2 wins by opting to move a neutral tile before moving their own. Conversely, if the other player starts, that player can only lose if they attempt to cover their opponent's tile with the "wrong" neutral tile (the one not adjacent to theirs). This situation suggests a general hypothesis that the player whose tiles have greater total adjacency to neutral tiles is at a disadvantage. Furthermore, in (D), where players have equal total adjacency to neutral tiles, P_1 simply cannot lose, suggesting, like (B), the preponderance of a turn-order advantage in cases of connective symmetry.

These examples demonstrate some fundamental patterns arising from movement constraints, relative mobility due to neutral tile topology, and small-scale (i.e., late game) board geometry, but they fail to capture the strategic depth enabled by High Tide's D6 dihedral symmetry. To study this aspect of the game in a tractable manner, we propose the **N-Tide** convention.

B. N-Tide

N-Tide is a variant of High Tide whose initial board position is a hex flower of radius N; more explicitly, for $N \geq 2$, the board contains a central yellow tile surrounded by $N-1$ complete hexagonal rings of tiles. The total number of tiles $H(N)$, and the total number of tiles of each color $P(N)$, $B(N)$, and $Y(N)$, are given by the following polynomials. Note that $H(N)$ is the N-th centered hexagonal number, which, in turn, equals 1 plus 6 times the $(N-1)$-th triangular number, giving an interesting expression for $\Phi_{max}(N)$.

$$P(N) = B(N) = N^2 - 2 \tag{9}$$

$$Y(N) = N^2 - 3N + 5 \tag{10}$$

$$\begin{aligned} H(N) &= 3N^2 - 3N + 1 = 3N(N-1) + 1 \\ H(N) &= 1 + 6\Big(\frac{N(N-1)}{2}\Big) \; = 1 + 6\, T(N-1) \end{aligned} \tag{11}$$

TABLE I
CANONICAL REPRESENTATIVE ELEMENTS AND ORBIT COUNTS OF EQUIVALENCE CLASSES OF INITIAL POSITIONS UNDER D6 DIHEDRAL SYMMETRIES IN 2-TIDE.

Class	Representative Element	Orbit Count
1	**Y** | **PPBBYY**	12
2	**Y** | **PPBYBY**	12
3	**Y** | **PBPBYY**	12
4	**Y** | **PBBYPY**	12
5	**Y** | **PPBYYB**	6
6	**Y** | **PPYBBY**	6
7	**Y** | **PBPYBY**	6
8	**Y** | **PBBPYY**	6
9	**Y** | **PBYPBY**	6
10	**Y** | **PBYPYB**	6
11	**Y** | **PBYBPY**	6

$$\Phi_{max}(N) = T(H(N)) = \frac{H(N)(H(N)+1)}{2} \tag{12}$$

For the original game ($N = 3$), "3-Tide", these numbers match the total counts of tiles available out of the box: **7P7B5Y**. For the $N = 2$ variant, "2-Tide", they likewise result in the inner ring as prescribed: some permutation of **2P2B2Y** plus the central **1Y**. The precise allocation of colors to the n-th layer, i.e., the $(n-1)$-th ring, obeys:

$$P_n = B_n = \begin{cases} 0, & n = 1 \\ 2, & n = 2 \\ 2n-1, & n \geq 3 \end{cases} \tag{13}$$

$$Y_n = \begin{cases} 1, & n = 1 \\ 2, & n = 2 \\ 2n-4, & n \geq 3 \end{cases} \tag{14}$$

By this convention, the fraction of tiles which are neutral, $\frac{Y(N)}{H(N)}$, approaches $\frac{1}{3}$ as N becomes large. We deem this to be an acceptable deviation from the original value $\frac{Y(3)}{H(3)} \approx 0.26$, as it would plausibly maintain a similar level of competitive tension.

C. 2-Tide

Whereas 3-Tide poses an immensely complex optimization problem, 2-Tide is far more tractable, with a mere $H(2) = 7$ tiles and thus a naive configuration space size of $\binom{7+7-1}{7} = \binom{13}{7} = 1,716$ weak compositions. Among these, as previously stated, the number of possible initial positions is $\binom{6}{2}\binom{4}{2} = 90$; furthermore, these positions comprise only 11 equivalence classes under rotation and reflection, here canonicalized in the form **Y** | **PPBBYY**, where the symbol on the left represents the color of the central tile and the string on the right represents the colors of petal tiles read cyclically with arbitrary starting cell index. The orbit counts of the equivalence classes, calculated by applying Burnside's lemma—see [5]—are displayed in Table I, and the corresponding representative elements are displayed in Figure 2.

IV. METHODS

A. Dynamic Programming

Because 2-Tide has only 11 canonical initial positions, it is relatively simple to compute a tablebase of play outcomes. To do this, we implement a retrograde minimax solver which uses a lexicographic score tuple σ to determine the winner in a terminal position. The first score element is ΔA, the difference between the apical tile counts for players p and q, where p is the maximizer and q is the minimizer.

$$\Delta A = \alpha_p - \alpha_q \tag{15}$$

Next is ΔH, the difference between the height of player p's tallest apical tall h_p^{max} and that of player q, h_q^{max}.

$$\Delta H = h_p^{max} - h_q^{max} \tag{16}$$

The third score element ΔL indicates by its sign which player moved last.

$$\Delta L = \begin{cases} +1, & p \text{ last} \\ -1, & q \text{ last} \end{cases} \tag{17}$$

Thus, the objective of p is to maximize each element of σ, while the objective of q is to minimize each element of σ.

$$\sigma = (\Delta A, \Delta H, \Delta L) \tag{18}$$

For position s, we define a winning function $V(s)$ which indicates the winning player among players p and q. If $V(s) > 0$, then the winner is p; else, if $V(s) < 0$ and s is a terminal position, then the winner is q.

$$V(s) = \begin{cases} +1, & (\Delta A > 0) \textit{ or } (\Delta A = 0 \;\&\; \Delta H > 0) \\ & \textit{or } (\Delta A = 0 \;\&\; \Delta H = 0 \;\&\; \Delta L > 0) \\ -1, & (\Delta A < 0) \textit{ or } (\Delta A = 0 \;\&\; \Delta H < 0) \\ & \textit{or } (\Delta A = 0 \;\&\; \Delta H = 0 \;\&\; \Delta L < 0) \end{cases} \tag{19}$$

For a legal move set $\mathcal{M}(s,p)$, let $\mathcal{M}^*(s,p) \subseteq \mathcal{M}(s,p)$ denote the subset of moves which are optimal. An *optimal move* $m^* \in \mathcal{M}^*(s,p) \mid m^* : s \to s'$ is one for which the fraction of terminal positions stemming from s' in which p is the winner is not less than that of any other move $m \in \mathcal{M}(s,p)$.

This framework enables us to deterministically compute the probability that p will eventually win starting from position s where p is the active player using dynamic programming (DP) with an ε-greedy move selection policy, described in [6]. Assuming player p picks an optimal move with uniform probability ε_p or any legal move with uniform probability $1 - \varepsilon_p$ (note that this inverts Sutton and Barto's convention), and q likewise picks an optimal move with uniform probability ε_q or any legal move with uniform probability $1-\varepsilon_q$, the probability that p wins eventually is determined by the following recursive piecewise function. Division by zero when $\mathcal{M}(s,p) = \emptyset$ (and thus $|\mathcal{M}(s,p)| = 0$) is prevented by the fourth case.

$$\varepsilon(s) = \begin{cases} \varepsilon_p, & \text{turn}(s) = p \\ \varepsilon_q, & \text{turn}(s) = q \end{cases} \tag{20}$$

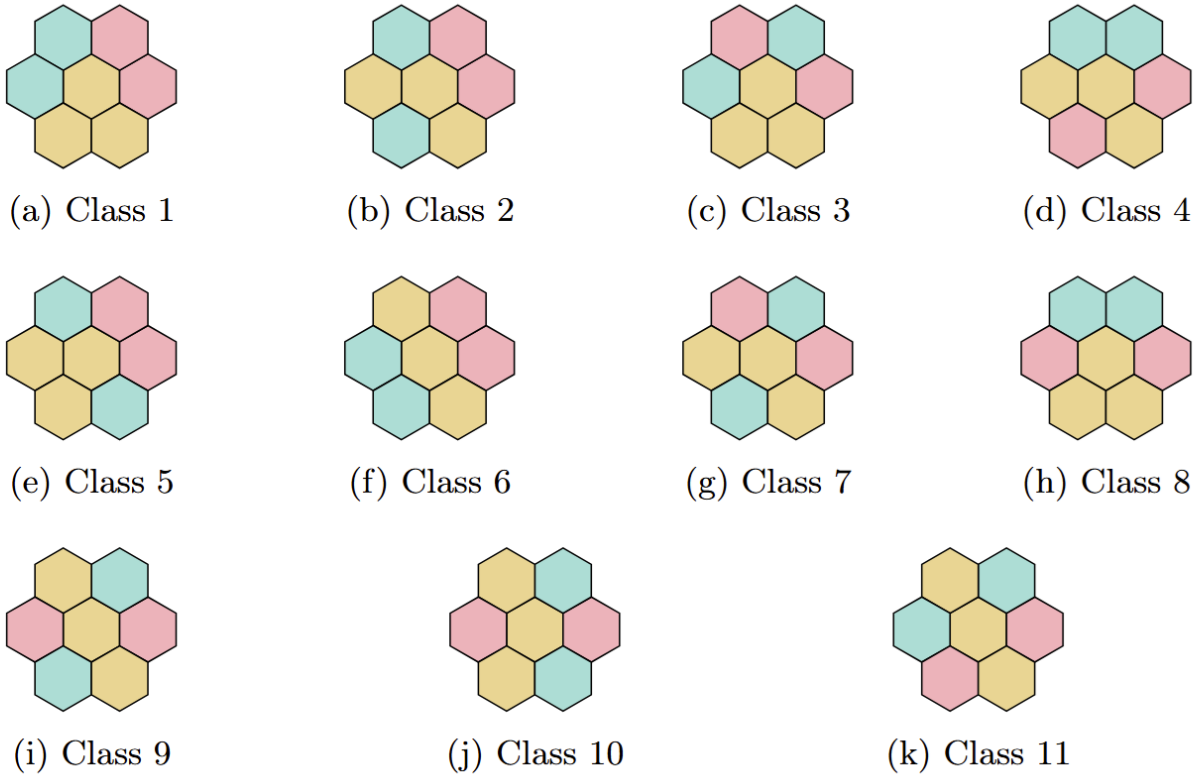


Fig. 2. The 11 canonical configurations of the inner ring for 2-Tide. Each subfigure represents one equivalence class under dihedral symmetry. Color inverse pairs include Class 2 ≅ Class 4, Class 5 ≅ Class 8, and Class 7 ≅ Class 10. Classes which are their own color inverse by reflection symmetry include Classes 1, 3, and 9. Classes which are their own color inverse by reflection symmetry *or* rotation symmetry include Classes 6 and 11.

$$Pr(p \text{ wins}|s,\varepsilon) = \begin{cases} 1, & |\mathcal{M}(s,p)| = 0 \;\&\; V(s) > 0 \\ 0, & |\mathcal{M}(s,p)| = 0 \;\&\; V(s) < 0 \\ W(s,p,\varepsilon), & |\mathcal{M}(s,p)| > 0 \;\&\; |\mathcal{M}^*(s,p)| > 0 \\ W'(s,p,\varepsilon), & |\mathcal{M}(s,p)| > 0 \;\&\; |\mathcal{M}^*(s,p)| = 0 \end{cases} \tag{21}$$

$$W(s,p,\varepsilon) = \varepsilon \cdot \frac{\sum_{m^* \in \mathcal{M}^*(s,p)} Pr(p \text{ wins}|m^*(s),\varepsilon)}{|\mathcal{M}^*(s,p)|} + (1-\varepsilon) \cdot \frac{\sum_{m \in \mathcal{M}(s,p) \setminus \mathcal{M}^*(s,p)} Pr(p \text{ wins}|m(s),\varepsilon)}{|\mathcal{M}(s,p) \setminus \mathcal{M}^*(s,p)|} \tag{22}$$

$$W'(s,p,\varepsilon) = \frac{1}{|\mathcal{M}(s,p)|} \sum_{m \in \mathcal{M}(s,p)} Pr(p \text{ wins}|m(s),\varepsilon) \tag{23}$$

One attractive strategy, given sufficient time and computer memory, would be to calculate and memoize win probabilities for all possible successor positions s' for $\varepsilon = 1$ and choose the move m which maximizes $W(s', p, 1)$. However, the explosion of position counts—and thus potential recursion depth—for $N \geq 3$ makes the computation of these probabilities infeasible except in positions which are at most a small number (relative to the maximum number) of degrees of separation away from terminal positions. Therefore, an alternative methodology was chosen for simulating games of the original High Tide board family, 3-Tide.

B. Monte Carlo Tree Search

Another powerful technique for optimal play of complex games is Monte Carlo Tree Search (MCTS), such as the Upper Confidence bounds applied to Trees (UCT) algorithm described in [7] and [4]. This algorithm has the advantage of great speed compared to exhaustive state enumeration, as only a small subset of the possible game tree is traversed stochastically on every turn. MCTS was implemented for simulations of 3-Tide with semi-randomized boards seeded from the canonical 2-Tide boards. For each 2-Tide board, 100 new 3-Tide boards were generated by adding to it an independently randomized outer ring of tiles. These represented minuscule samples of the space of all possible initial boards, which as previously shown would be unrealistic to enumerate at the scale of this study. However, organizing boards by interior canonical class in this way enabled a comparison between 2-Tide and 3-Tide with the goal of elucidating whether the board interior has any measurable effect on the outcomes of games.

Traces were collected from ∼550 simulated games (totaling 605,549) played on each of these boards with both players using an MCTS decision scheme with a UCT exploration constant $c = 1.4$. On each turn, 400 Monte Carlo rollout iterations (up to a turn limit of 20,000 iterations) were

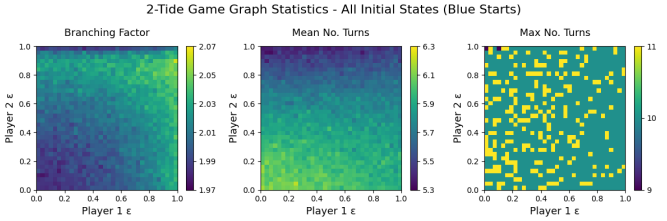


Fig. 3. Game graph statistics derived from 2-Tide simulations over ε-greedy policy combinations. *Left:* Branching factor (mean number of legal moves per node), player-agnostic. *Middle:* Mean number of turns in a game. *Right:* Maximum number of turns in a game.

simulated from each available legal move, and wins were tallied to calculate a win probability for each move. Pilot runs over a range of numbers of per-move iterations showed 400 to be an acceptable trade-off between Monte Carlo noise and run time. The (player-specific) win probabilities were cached and re-used on successive turns. During roll-out move selection, if any legal moves resulted in a terminal state with the active player winning, one of them was chosen uniformly randomly; otherwise, if any resulted in a terminal state with the active player losing, blunders were avoided by choosing a uniform random move from among all legal moves except those.

V. Results

A. 2-Tide Simulations

The set of ε-greedy 2-Tide simulations covered $34 \times 34 = 1,156$ combinations of $\varepsilon_1, \varepsilon_2$ values in steps of ~ 0.0303 between 0.0 and 1.0, inclusive, including 10,000 runs per combination, for each of the 11 canonical initial positions, for both blue and pink as the starting player, totaling 254,320,000 games. Game graph statistics appear in Figure 3. Mean branching factor b, the mean number of legal moves available in every position (including terminal positions), ranged between 1.97 and 2.07, varying linearly with the root sum of squares $\sqrt{\varepsilon_1^2 + \varepsilon_2^2}$. This reflected the fact that suboptimal play results in more blunders. Additionally, it shows that for 2-Tide, a binary tree ($b = 2$) models the game graph topology with surprising accuracy. The mean number of turns per game ranged from $\sim$5.3 to $\sim$6.3, remaining constant with respect to ε_1 but decreasing approximately linearly with increasing ε_2. The maximum number of turns was 10 or 11 in all but two $\varepsilon_1, \varepsilon_2$ conditions, with no apparent correlation to either ε.

Figures 4 and 5 show the P_2 win rate over $\varepsilon_1, \varepsilon_2$ combinations, with blue and pink as the respective starting player in each figure. Where blue is the starting player, low P_2 win rates are colored in blue and high P_2 win rates are colored in red; vice-versa where pink is the starting player. Black dotted lines on heatmaps and dashed lines on line plots indicate $\varepsilon_1 = \varepsilon_2$. The overall trends are *nearly* the same across the 11 canonical initial positions regardless of which color is P_1, with four of the positions favoring P_2 in matched optimal play and the remaining seven favoring P_1. In both cases, the positions favoring P_2 include Classes 1, 6, and 11 in Figure 2. However, when blue starts, Class 2 favors P_2 and Class 4 favors P_1, yet when pink starts, these are reversed: Class 2 favors P_1 and

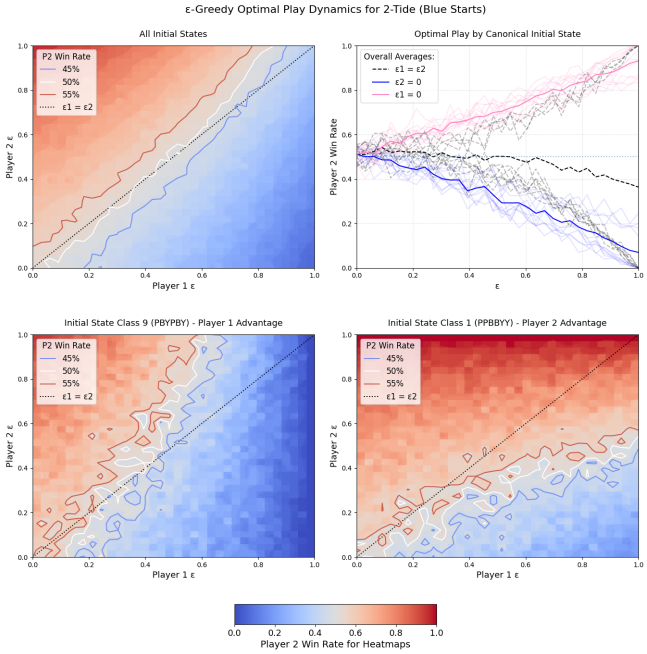


Fig. 4. P_2 win probabilities (when Blue starts) across 10,000 simulations per combination of ε-greedy play optimization values in 2-Tide. 1,156 combinations were tested for each of the 11 canonical initial positions (CIP). *Upper Left:* Heatmap of win rates over ε_1,ε_2 combinations for all CIP. White line is the 50% win rate contour, red line is 55%, and blue line is 45%. Faint shaded bands are win rate contours in increments of 5%. Dotted black line indicates equal ε. *Upper Right:* P_2 win probability as a function of ε_1 when $\varepsilon_2 = 0$ (blue), as a function of ε_2 when $\varepsilon_1 = 0$ (pink), and as a function of $\varepsilon = \varepsilon_1 = \varepsilon_2$ (dashed). For four out of eleven CIP—Classes 1, 2/4, 6, and 11—matched fully optimal play resulted in a P_2 win (blue/pink starting) in nearly 100% of simulations, and for the other seven—classes 3 through 5 and 7 through 10—a P_1 win. This indicates that outcomes depend on initial board geometry. *Lower Left:* Heatmap for a CIP with significant P_1 advantage (Class 9, PBYPBY). *Lower Right:* Heatmap for a CIP with significant second-player advantage (Class 1, PPBBYY).

Class 4 favors P_2. That is, Class 2 always favors pink and Class 4 always favors blue. The favored players for all classes are summarized in Table II.

To address the key differences between the classes favoring P_2 and those favoring P_1, we created log files to record the trajectories of moves chosen in all optimal-optimal games and compared their motifs.

a) Center Stacking: In P_2-favored class trajectories, 34.9% of opening moves by P_1 landed on the center tile, compared to 2.8% for P_1-favored classes. P_2 win rate rose significantly when this occurred; in particular, from 68.7% and 73.1% in Classes 2 and 6 respectively when the opening move did not land in the center, both to 98.5% when it did.

Play also tends toward the center overall throughout games in P_2-favored classes, with a mean number of 3.64 moves destined for the center cell per game compared to 1.90 moves per game in the P_1-favored classes.

b) Early Neutral Tiles: Whereas in P_1-favored class trajectories the first move by P_2 was fairly evenly split between tiles of their own color (54%) and neutral tiles (46%), the P_2-favored classes saw 77% of P_2's first moves use a neutral tile.

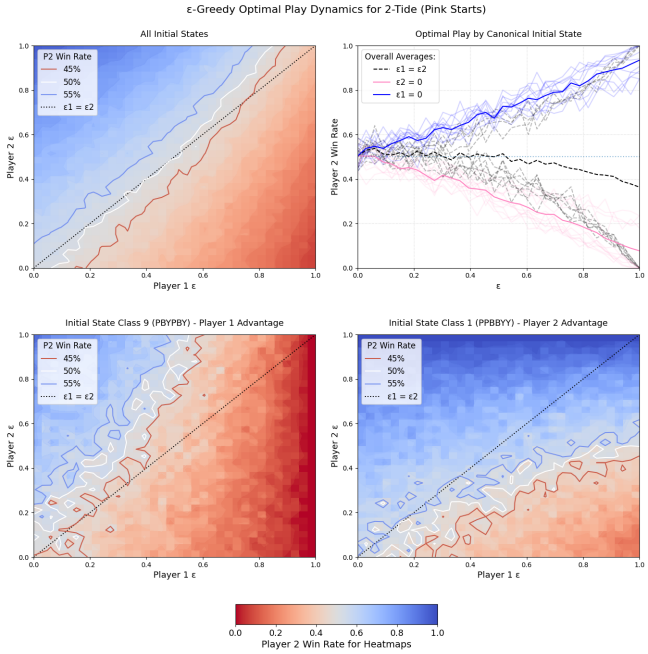


Fig. 5. P_2 win probabilities when Pink starts for otherwise identical parameters to those resulting in Figure 4, including the same random seed value 7. Overall trends are nearly identical, or perfectly identical in the case of $\varepsilon_1 = \varepsilon_2$, and equivalence classes have polarized win rates in the same ways for two optimal players regardless of which color starts, with the sole exception of Classes 2 and 4.

TABLE II
WINNING PLAYERS IN MATCHED OPTIMAL PLAY OF 2-TIDE CANONICAL CLASSES BY A DYNAMIC PROGRAMMING DECISION SCHEME, AS WELL AS P_2 WIN RATES IN MCTS APPROXIMATION OF OPTIMAL PLAY OF 3-TIDE FOR 100 BOARDS PER CLASS, ∼550 GAMES PER BOARD.

Class	Rep. Element	2-Tide Winner (DP)	3-Tide P_2 Win Rate
1	**Y | PPBBYY**	P_2	0.56
2	**Y | PPBYBY**	P_2 *(Blue start)* P_1 *(Pink start)*	0.45
3	**Y | PBPBYY**	P_1	0.47
4	**Y | PBBYPY**	P_1 *(Blue start)* P_2 *(Pink start)*	0.54
5	**Y | PPBYYB**	P_1	0.47
6	**Y | PPYBBY**	P_2	0.42
7	**Y | PBPYBY**	P_1	0.59
8	**Y | PBBPYY**	P_1	0.48
9	**Y | PBYPBY**	P_1	0.55
10	**Y | PBYPYB**	P_1	0.49
11	**Y | PBYBPY**	P_2	0.44

B. 3-Tide Simulations

Play sequence traces from MCTS 3-Tide simulations did not significantly recapitulate P_1 and P_2 advantages specific to the 2-Tide canonical classes of the board interior. P_2 win rates (Table II) ranged only from 42%-59% across these classes, and the P_1-favored classes in 3-Tide were not the same as in 2-Tide (and vice-versa). For example, whereas Class 6 and Class 7 favored P_2 and P_1 respectively in 2-Tide, they exhibited the lowest (Class 6) and highest (Class 7) P_2 win rate in 3-Tide. Overall, P_1 won 50.36% of all games and P_2 won 49.64%.

a) Mobility and Self-Covering: The strongest predictors of the winner were the number of moves available on each player's turn and the rates at which players selected moves that covered their own tiles. P_2 win rate was correlated with the fraction of P_1's moves that covered P_1's own tiles ($r = 0.2179,\ p < 0.0001$) and anticorrelated with the fraction of P_2's moves that covered P_2's own tiles ($r = -0.2311,\ p < 0.0001$). The trend for mobility appears at first counterintuitive: P_2 win rate was correlated *positively* with the mean number of moves available to P_1 on P_1's turns ($r = 0.1984,\ p < 0.0001$) and *anticorrelated* with the mean number of moves available to P_2 themself ($r = -0.2340,\ p < 0.0001$). An explanation of this phenomenon is proposed in Section VI.

b) Covering Enemy Tiles: Another important factor was the overall fraction of each player's moves that covered their opponent's tiles. The P_2 win rate was moderately correlated ($r = 0.1955,\ p < 0.0001$) with this number for P_2 (covering P_1's tiles) and anticorrelated ($r = -0.1737,\ p < 0.0001$) with the number for P_1 (covering P_2's tiles). This was an aggregate pattern spanning whole games; the correlation with the number of times P_2 covered P_1's tiles in the early game (within P_2's first 3 turns) was much smaller but still significant ($r = 0.0594,\ p = 0.0487$), while there was no such correlation for P_1 ($r = -0.0198,\ p = 0.5111$), perhaps because P_2's chosen response to the opening move, operating on a narrowed game tree, may differ more greatly in expected win probability from the other options than the opening move differed from its siblings.

c) Neutral Tiles: Contrary to the 2-Tide result, there was no correlation between either P_1 or P_2 moving a neutral tile on their first turn and either player winning ($r = 0.0182,\ p = 0.5472$ for P_1 opening by moving a neutral tile, $r = -0.0226,\ p = 0.4535$ for P_2 moving a neutral tile as the first reply).

As for initial board topology, mild but statistically significant anticorrelations existed between P_2 win rate and the number of edges between P_2 tiles and neutral tiles within the outer ring ($r = -0.0897,\ p = 0.0029$) and the mean fraction of all tiles neighboring outer ring P_2 tiles which were neutral ($r = -0.1019,\ p = 0.0007$).

VI. DISCUSSION

Taken together, the 2-Tide statistics suggest that, like in the "Micro-Tide" variants explored previously, P_2's capacity to win in 2-Tide depends strongly on the availability of opening replies that preserve the mobility of P_2's own tiles. If P_1's first move leaves an opportune move of a neutral tile, P_2 can take it and force a cascade of moves into the center that they will finish.

The topological dependence of the polarization of 2-Tide outcomes is not obvious. Board classes which are their own color inverses under certain symmetries include Classes 1, 3, 6, 9, and 11. Among these, Classes 1, 6, and 11 favor P_2, while Classes 3 and 9 favor P_1. This non-uniformity confirms the inapplicability of Nash's strategy-stealing argument, which would otherwise imply that on boards such as these, where

P_1's strategy is color-invariant, all would favor P_1, since either player would make the same first move. Another puzzling observation is that the fourth class to favor P_2 when blue starts, Class 2, and the one favoring P_2 when pink starts, Class 4, are a color inverse pair. Meanwhile, Classes 5 and 8 and Classes 7 and 10 are also color inverse pairs, yet they all favor P_1. In Classes 7 and 10, there are equal numbers of edges between movable neutral tiles and either player's tiles; this is also true for all of the self-inverse classes. In Classes 2 and 4, the losing color (blue and pink respectively) has three edges with movable neutral tiles, whereas the winning color has only one. This would suggest a simple rule that the difference in initial neutral tile adjacency predicts the winner, in agreement with the correlation found in 3-Tide—but that would be foiled by the fact that both Classes 5 and 8 always favor P_1, despite a different color in each having no edges to movable neutral tiles at all. Here we see that the effects of turn order and initial board asymmetry are separable to some extent, but the solution to this paradox lies deeper in the game's dynamics.

The 3-Tide traces help clarify the connection between board topology, movement patterns, and win probabilities. While counterintuitive, the anticorrelation between a player's abundance of moves and their win rate can be seen as a consequence of the movement constraints. Because tiles can only be moved upward, an abundance of moves is correlated with one's tiles being scattered at low stack heights rather than occupying valuable high vertical positions. Moreover, because moves must not split the board into disconnected subgraphs, these positions are often shielded by a pinned tile rather than being exposed, as most low tiles are until the late game.

Furthermore, rather than being determined independently by the board interior, the outcome of 3-Tide is influenced by color assortativity of exterior tiles, which depends both on exterior-exterior and exterior-interior connections. Whereas in 2-Tide the difference in neutral tile adjacency was not always predictive of the winner, 3-Tide results indicate that this was likely because the assortativity effect was curtailed by all 2-Tide ring tiles being movable. In 3-Tide, inner ring tiles cannot be moved to preemptively avoid their own covering, and high neutral tile adjacency has three main consequences. First, adjacency of a player's exterior tiles to inner ring neutral tiles encourages lateral rather than inward moves in order to cover their opponent's tiles, leading to pinned stacks in the periphery. Second, adjacency of a player's inner ring tiles to neutral exterior tiles significantly increases the likelihood that the opponent will cover them with those neutral tiles to protect their own, leading to interior stacks that are more fluid in the mid-game. And third, high adjacency to neutral tiles often implies low adjacency between one's own tiles, in which case the option of an immediate counter-cover—reminiscent of *en passant*—is less likely.

VII. Conclusion

While reduced versions of the game are prone to strategy-stealing behavior and outcomes that are overdetermined by the initial board topology, the original High Tide board is sufficiently complex to resist absolute domination by a win probability-maximizing decision scheme for a large number of tested initial positions. However, motifs from optimal play at both scales lend some actionable insight (beyond the obvious principles of trying to cover the opponent's tiles and avoiding self-occlusions). First, neutral tiles can be a powerful defensive resource if they are used to cover the opponent's tiles (preempting the converse move by the opponent), but aggregate neutral tile use is not a decisive factor for victory. The second player may be at a considerable disadvantage if their tiles have greater initial adjacency to neutral tiles than the first player's, because this will give the first player more ways to cover the second player's tiles than the second player has to cover the first player's. This and any other situation in which either player can critically reduce their opponent's non-self-occlusive option set can significantly shorten games by creating a Zugzwang-like position. In general, one should avoid covering their own tiles and conflating availability of moves with strength of board position. To counter the mild-to-moderate first player advantage (which depends on initial board topology), the second player should focus on covering enemy tiles in the first few turns, and use neutral tiles adjacent to their own to do so if able.

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